

## Using IT to make buildings safer

Purdue University (USA) engineers, computer and graphics technology experts simulated the ill-fated Boeing 757 crash into the Pentagon on September 11, 2001. Scientific principles have been used to understand what really happened in theory. The simulation combines a realistic-looking visualisation of the airline approaching the building with a technical, science based animation of the plane crashing into the structure.

What is distinctive about the software used is that it uses principles of Physics to simulate the impact of an airline's massive fuel and cargo on a building.

To start with, computers were used to create a mathematical model of reinforced concrete columns. This was then used as an origin to produce the simulation. Thereafter, this model was turned into the simulation by expressing the plane and its weight as a mesh of hundreds of thousands of finite elements.

This mesh requires millions of calculations per second of simulation. Creating just one-tenth of a second of simulation took as many as 95 hours of a supercomputer's computing time.

Hence, this study has enabled us to cross a huge obstacle of merging commercial software (one that is normally used by auto manufacturers to simulate car crashes) with Maths based models that enable simulation of a plane crashing into a building. Findings of such simulation are of high utility to civil engineers who are involved with the design of reinforced concrete structures (especially critical buildings like ammunition centres and hospitals) that need to be protected more from such fatalities.

## One World Trade Centre

As we all are aware, on September 11, 2001, a massive terrorist attack

destroyed the twin towers of World Trade Center in lower Manhattan, New York, USA. In the recovery phase, a number of designs were submitted for reconstruction and finally architectural firm Skidmore, Owings & Merrill LLP's was selected.



Fig. 6: Freedom Tower, New York, USA (Image courtesy: nyposter.com)



Fig. 7: A church made by using self-cleaning concrete (Image courtesy: www.nanowerk.com)

One World Trade Center has been dressed in sparkling glass with the overall design reflecting Statue of Liberty. Its distinctive structural system includes a cable-net structure that looks like Brooklyn Bridge.

Autodesk, the design and drafting software expert firm, was used for the information technology inputs by the architectural firm. Building Maker tools in Autodesk Revit 7 facilitated a thorough understand-

ing of the association between the tower's expressive and built forms, making the transition from design concept to construction-related finer details more supple.

Autodesk's AutoCAD flagship app was the project's primary 2D and 3D design and documentation tool. It enabled a large number of engineers to work together in parallel and communicate on real-time basis in a 3D environment.

## Future building blocks

When we give more control of the design process over to the computer, the very speed and precision of the machines cut down the tedious process of exploration. This enables architects to create more inspiring and meaningful designs.

Artificial intelligence is being used to study structural designs. An intelligent computer-assisted design system (ICADS) is a knowledge based building design with expert agents. Expert agents based on artificial intelligence principles support the architect in the development, analysis, synthesis and evaluation of design solutions during initial phases. They have domain-specific knowledge. For example, an architect might want to use expert systems for controlling noise insulation or heat conservation.

To help further, we have nanotechnology, which is an area of science and technology that pertains to dimensions and tolerances in the range of 0.1nm to 100nm. It is being increasingly applied in construction materials research and development. Interestingly, we already have many building materials that are based on nanotechnology.

As a result, there are self-cleaning windows, self-healing concrete, materials to block ultraviolet and infrared radiation, smog-eating coatings and even light-emitting walls and ceilings. What we imagined as science fiction once is turning into reality today! **EFY**